MIL-P-19131(SHIPS)
AMENDMENT-4
20 January 1964
SUPERSEDING 1
Amendment-3
1 October 1962

MILITARY SPECIFICATION

PUMPS, ROTARY, POWER DRIVEN, MISCELLANEOUS

This amendment forms a part of Military Specification MIL-P-19131 (SHIPS), 23 January 1956.

Page 1, paragraph 1.2: Add:
"Type II - Screws with timing gears.
"Type X - Vane (sliding)."

Pages 1, 2, 3 and 4, paragraph 2.1: Delete reference to Military Specifications "MIL-T-15071", "MIL-N-17506", 'NAVY DEPARTMENT General Specifications for Inspection of Material" and "General Specifications for Ships: Section S9-1 Welding and Allied Processes.", and add the following:

"SPECIFICATIONS

"FEDERAL
"QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought (K-Monel).

"MILITARY
"MIL-Q-9858 - Quality Control System Requirements"
"MIL-M-15071 - Manuals, Equipment and Systems."

"STANDARDS

"MILITARY
"MIL-STD-278 - Welding and Allied Processes for Machinery for Ships of the United States Navy."

"PUBLICATIONS

"BUREAU OF SHIPS
"NAVSHIPS 250-637-2 - Instructions for Torch Brazing of Ferrous and Non-Ferrous Piping.
"NAVSHIPS 250-648-8 - Inspection and Test of Silver-Brazed Piping Systems."

Page 5, table I, column 5: Delete "33 - 100" and substitute "30 - 100".

Page 6, paragraph 3.3.1 item (e): Delete and substitute:

"(e) Satisfactory operation when inclined as follows, unless otherwise specified in the contract or order:
(1) Up to 30 degrees from the normal horizontal position in any direction (permanently inclined).
(2) With the ship rolling up to 60 degrees from the vertical (time of complete roll cycle, 8 seconds).

1/ CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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(3) With the ship pitching 10 degrees up and down from the normal horizontal plane (time of complete pitch cycle, 4 seconds)."

Page 14: Add as paragraph 3.3.15.3:

"3.3.15.3 "Horizontal flexibly coupled units with bedplates shall have coupling guards provided. Guards shall be designed to permit ready access to the coupling for lubrication and inspection."

Page 14: Add as paragraph 3.3.16.7:

"3.3.16.7 When specified (see 6.1) pumps shall be provided with mechanical seals."

Page 15, paragraph 3.3.19: Delete and substitute:

"3.3.19 Welding. - Welding shall be in accordance with Standard MIL-STD-278."

"3.3.19.1 Silver brazed joints shall be in accordance with NAVSHIPS 250-637-2. For joints 1/2 inch and above, only preinserted ring fittings shall be used.

Page 16, table II, column 3, line 7: Delete "MIL-N-17506" and substitute "QQ-N-286".

Page 17, paragraph 3.4.3.1: Delete and substitute:

"3.4.3.1 Materials for class N-4 and 0-3 pumps shall be the same as listed in table II except that commercial materials may be used in place of Government specification materials."

Page 17: Add as paragraph 3.4.3.3:

"3.4.3.3 Materials of the vanes of types IX and X pumps shall be that most suitable for classes N-2 and 0-2 services and shall be as proposed by the manufacturer and as approved by the bureau or agency concerned."

Page 18, paragraph 3.5.3.1: Delete formula "$1.5 = (\frac{\text{above formula hp.}}{4})$" and substitute:

"$1.5 (\frac{\text{above formula hp.}}{4})$"

Page 21, paragraph 3.8.1: Delete and substitute:

"3.8.1 Manuals shall be in accordance with Specification MIL-M-15071."

Page 21, paragraph 3.8.5, line 5: Delete "MIL-T-15071" and substitute "MIL-M-15071".

Page 22, table III: Delete and substitute:

"Table III - Onboard repair parts

Item	Name of part	Quantity per set
a b c d e	Ball or roller bearings Oil seals Packing, if special, and packing spacers Coupling bushings, pins, springs, and other wearing parts Mechanical seals	100 percent complete replacement for all installed units per ship

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"Table III - Onboard repair parts (continued)

Item	Name of part	Quantity per set
f	Sleeve, sliding surface bearings	One to four units per ship
g	Cams, slides, slide pins, housings or liners, vanes (set) and similar wearing parts	100 percent replacement for one pump Over four units per ship 100 percent replacement for two pumps
h	Timing gears, reduction gears and pinions	100 percent replacement for one pump
i	Complete rotating element including bearings and complete coupling	

Page 22, paragraph 3.9.2.1, line 1: Delete "item (f)" and substitute "item (g)".

Page 23, paragraph 3.9.2.2, line 1: Delete "item (h)" and substitute "item (i)".

Page 23, paragraph 3.9.2.3, lines 1 and 2: Delete "items (a) and (g)" and substitute "items (a) and (h)", and in line 3, delete "item (h)" and substitute "item (i)".

Page 25, paragraph 4.2.1: Delete and substitute:

"4.2.1 Qualification tests. - Qualification tests on class N-2 and class O-2 pumps shall be conducted at a laboratory satisfactory to the Bureau of Ships. These tests shall consist of the test specified in 4.2.8.

Page 25, paragraph 4.2.2: Delete and substitute:

"4.2.2 In addition to the tests at the laboratory class N-2 and O-2 pumps may be required to pass the following shipboard service test:

(a) Two or more pumps of the same general design and general operating characteristics shall have given satisfactory service onboard Naval ships for a period of at least two years and shall still be in service on such ships at the time qualification approval is granted."

Page 27, paragraph 4.3.2, lines 1 and 2: Delete "as selected by the Government inspector".

Page 28, paragraph 4.3.3.1: Delete class "O-3" and substitute class "O-2"; line 2, delete "tested on order" and substitute "tested in order".

Page 28, paragraph 4.3.3.2, line 1: Delete class "H-2" and substitute class "N-2".

Page 28, paragraph 4.3.4.1, line 2: Delete "by the representative of the bureau or agency concerned."

Page 30, paragraph 4.3.5, first sentence: Delete and substitute "Sufficient data shall be taken during the tests of 4.3.1, 4.3.2 and 4.3.3 to prepare pump characteristic curves as specified below."

Page 33, paragraph 4.5: Delete and substitute:

"Inspection and test of silver brazed joints in systems designated as hazardous in manual NAVSHIPS 250-648-8 shall be in accordance with the requirements of that manual."

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Page 33: Add as paragraph 4.6:

"4.6 QUALITY ASSURANCE PROVISIONS

"4.6.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

"4.6.1.1 Quality control system. - The supplier shall provide and maintain a quality control system acceptable to the Government. The system of quality control shall be in accordance with MIL-Q-9858 and as specified in 4.5.1.1.2 through 4.5.1.1.5. The supplier is in no way relieved of the final responsibility to furnish pumps meeting the requirements of this specification.

"4.6.1.2 Description of procedures. - Procedures shall be assembled in manual form and shall indicate organization and responsibility for the control of quality.

"4.6.1.3 Inspection during manufacture. - The suppliers shall establish and maintain inspection at appropriately located points in the manufacturing process, beginning with the receipt of raw materials and subcontracted parts through fabrication, assembly, testing and preparation for shipment of completed units, to assure continuous control of quality of parts, components, assemblies, and construction including welding and fabrication.

"4.6.1.4 All instruments used in performance acceptance tests shall be calibrated before and after tests.

"4.6.1.5 No instruments containing mercury shall be used in the manufacture or testing of any equipment destined for installation on a nuclear powered ship. This point shall be covered in the manufacturer's quality control manual. Test reports shall contain a certification signed by a responsible official of the manufacturer that no mercury containing instruments have been used in the manufacture or testing of the equipment."

Page 35, paragraph 6.1 item (h)(9): Delete and substitute:

"(h)(9) Whether mechanical shaft seals are required (see 3.3.16.7)."

Preparing activity:
Navy - Ships
(Project 4320-N088Sh)

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